

Single-Photon Avalanche Diode Model for Circuit Simulations

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Abstract—In this letter, we present a detailed circuit model for single-photon avalanche diodes. The model can be implemented in many computer aided design circuit simulation environments and provides a representation of the detector's behavior in the above-breakdown (Geiger-mode) operation. The model implements the self-sustaining and self-quenching processes and correctly characterizes the reverse current–voltage curve. We show very good matching with experimental measurements obtained with both passive and active quenching circuits, and both single-photon and photon burst excitations.

Index Terms—Avalanche photodiodes, circuit simulation, modeling, single-photon avalanche diode (SPAD).

I. INTRODUCTION

SINGLE-PHOTON counting and timing applications have been developed over many years. Significant results have been reached with single-photon avalanche diodes (SPADs) in different areas such as quantum cryptography, astronomy, single molecule detection, fluorescent decays, luminescence measurements, etc. [1]. Essentially, SPADs are p-n junctions biased at a voltage V_A above the breakdown voltage V_B . At this bias, the electric field is so high that a single charge carrier injected in the depletion layer can trigger a self-sustaining avalanche. The current rises swiftly to a macroscopic steady level, in the milliamperes range. If the primary carrier is photogenerated, the leading edge of the avalanche pulse marks the arrival time of the detected photon with a precision of few tens of picoseconds. The current keeps flowing until the avalanche is quenched when the bias voltage is lowered down to or below V_B . Then the bias voltage must be restored, in order to be able to detect another photon. These operations require a suitable circuit, usually referred to as a quenching circuit [1].

In order to accurately simulate the behaviour of the quenching circuit during the design phase, a suitable model of the SPAD, capable of modeling the avalanche current ignition, quenching, and recovery processes, is required.

II. BASIC SPAD MODEL

The simplest quenching circuit is the commonly called passive quenching circuit (PQC), shown in Fig. 1 (left). The avalanche current is self-limited by the high-value ballast load

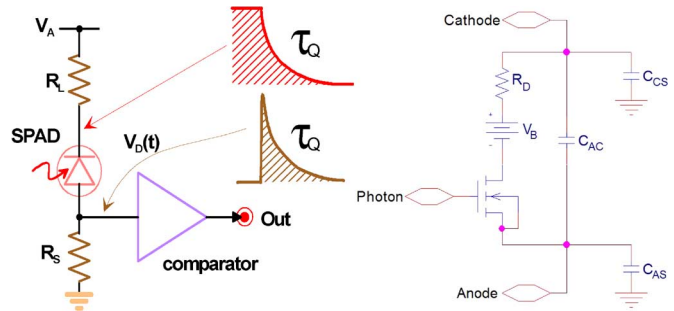


Fig. 1. Basic PQC (left) and traditional SPAD model (right). The transistor switch mimics the avalanche triggering.

R_L (about 100 k Ω or more). When no current is flowing, the SPAD cathode is reverse biased at V_A above breakdown; as soon as a photon fires the multiplication process, the avalanche current quickly rises to few milliamperes, generating a voltage drop across R_L , that eventually reduces the junction voltage. A fast discriminator senses the steep onset of the avalanche current on a small-value sense resistor R_S , connected to the anode [1].

In order to properly account for quenching and recovery times, some parasitic components must be considered, as in the traditional basic SPAD model (Fig. 1 right). C_{AC} is the junction capacitance and C_{AS} and C_{CS} are the stray capacitances from anode and cathode to substrate (typically few picofarads). The diode resistance R_D is given by the sum of the space-charge resistance and the resistance of neutral regions crossed by the avalanche current. V_B is the breakdown voltage and the transistor controls the avalanche triggering. When no current is flowing ($I_D = 0$), the diode bias voltage V_D equals V_A (the total reverse bias). Avalanche triggering is mimicked by closing the n-MOS switch. In few picoseconds, before V_D has changed, the current I_D reaches the peak value I_{PK} , then I_D and V_D exponentially decrease to the steady-state I_f and V_f values, respectively, given by [1]

$$I_{PK}(0^+) = \frac{V_A - V_B}{R_D} \quad I_f = \frac{V_A - V_B}{R_D + R_L} \quad V_f = V_B + R_D I_f \quad (1)$$

where $V_A - V_B$ is the excess bias above the breakdown voltage V_B . The quenching time constant τ_q is given by

$$\tau_q = (C_{AC} + C_{CS}) \cdot R_L // R_D \cong (C_{AC} + C_{CS}) \cdot R_D. \quad (2)$$

If I_f is very small, the current intensity becomes so low and the carriers that cross the avalanche region so few that it can happen that no carrier impact ionizes while crossing the high-field region. It is usually assumed that the avalanche process is self-sustaining if the current is higher than a threshold I_q of about

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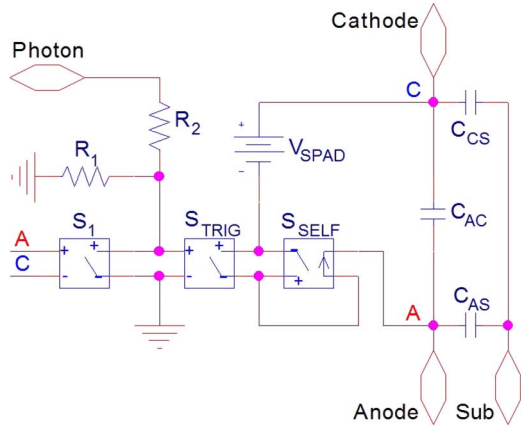


Fig. 2. Enhanced SPAD model. The “Photon” input ignites the detector through the switch S_{TRIG} . S_{SELF} self-sustains and self-quenches the avalanche process. V_{SPAD} describes the nonlinear $I-V$ curve.

100 μA [1]. Otherwise the process self-quenches, the switch in Fig. 1 gets open, and the two capacitors slowly recharge to V_A with a recovery time constant given by $\tau_r \approx R_L \cdot (C_{AC} + C_{CS})$. Now the transition, dominated by R_L , is very slow; for instance, with $R_D = 1$ k Ω , $R_L = 100$ k Ω , $C_{CS} = 1$ pF, and $C_{AC} = 9$ pF, we obtain $\tau_q = 10$ ns, while $\tau_r = 1$ μs .

It is well known that PQC suffers from severe limitations. For this reason, the active quenching circuit (AQC) was introduced: it senses the rise of the avalanche pulse and reacts back to the SPAD, by forcing the quenching and recovery transitions in short times (few nanoseconds) [2], [3]. For both passive and AQCs, either with discrete components or integrated approach, an improved model is needed, able to *self-sustain* and *self-quench* the avalanche process and to accurately describe the avalanche current waveform versus the excess bias.

III. ENHANCED SPAD MODEL

We conceived the model for solving all such issues, shown in Fig. 2. It is built by using standard components from the Orcad Cadence library and can be easily adapted to other circuit simulators. It simulates the *triggering*, the *self-sustaining* process and the *self-quenching* of the avalanche by means of current-voltage ($I-V$) controlled switches. The SPAD current and voltage are properly simulated by means of a voltage-dependent resistance and stray capacitances. The stray capacitances C_{CS} and C_{AS} are connected to the SPAD substrate, which is either left floating or is electrically connected to the metallic package.

Usually, the measured SPAD $I-V$ characteristic above breakdown is nonlinear, as shown in Fig. 3. Therefore, we decided to model it through a piecewise linear curve with different slopes, implemented through the nonlinear voltage generator V_{SPAD} shown in Fig. 2. Fig. 4 shows how to implement such a voltage-dependent generator into the property editor of Orcad PSpice, as a PSpice Template textual line. The user measures the $I-V$ characteristic, selects some (V_{Di}, I_{Di}) points along the experimental curve, possibly where the curve changes slope, and finally inputs them into the V_{SPAD} generator. Hence, the simulator computes the $R_a - R_e$ resistances of the SPAD at different excess biases,

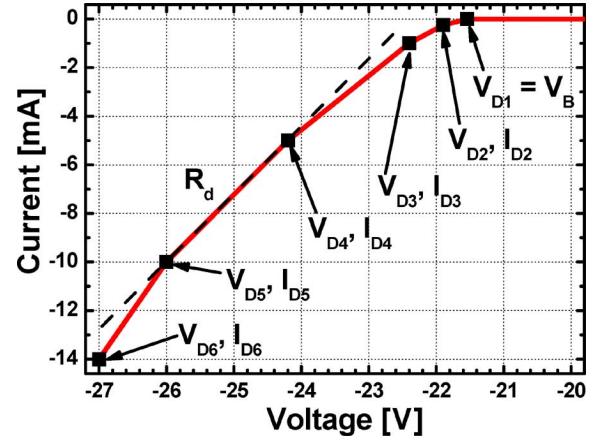


Fig. 3. Example of an SPAD [4] $I-V$ characteristic. The curve was measured by means of a curve tracer. Then the (V_{Di}, I_{Di}) coordinates where the slope changes were identified and were input into the V_{SPAD} generator of Fig. 2.

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PSpice Template: E^@REFDES %+ %- VALUE = {@Curve}
Curve: IF( V(%+, %-) < @VD1, @VD1, \n+
  IF( V(%+, %-) < @VD2, \n+
    @VD1 + @Ra * I (E^@REFDES), \n+
    IF( V(%+, %-) < @VD3, \n+
      @VD2 - @Rb * @ID2 + @Rb * I (E^@REFDES), \n+
      IF( V(%+, %-) < @VD4, \n+
        @VD3 - @Rc * @ID3 + @Rc * I (E^@REFDES), \n+
        IF( V(%+, %-) < @VD5, \n+
          @VD4 - @Rd * @ID4 + @Rd * I (E^@REFDES), \n+
          @VD5 - @Re * @ID5 + @Re * I (E^@REFDES) ) ) ) ) )
Ra: (@VD2 - @VD1) / (@ID2)
Rb: (@VD3 - @VD2) / (@ID3 - @ID2)
Rc: (@VD4 - @VD3) / (@ID4 - @ID3)
Rd: (@VD5 - @VD4) / (@ID5 - @ID4)
Re: (@VD6 - @VD5) / (@ID6 - @ID5)
VD1-VD6 and ID1-ID6: values input by the user

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Fig. 4. Implementation of the V_{SPAD} voltage source of Fig. 2 into the property editor of Orcad PSpice by Cadence.

thanks to the “Curve” computation in Fig. 4, which selects the correct range by checking the instantaneous V_D voltage across the SPAD and providing the corresponding resistance.

In quiescence, the voltage-controlled switch S_{TRIG} and the current-controlled switch S_{SELF} are open and the SPAD voltage V_D is the whole reverse voltage V_A . The “Photon” input is driven by a short positive pulse that acts as the avalanche trigger. When the positive input of S_{TRIG} exceeds a threshold level (a user-defined property of the switch), the switch closes and mimics the avalanche triggering. The current steeply rises to few milliamperes, thus exceeding the threshold level of S_{SELF} , set to I_q . Even if S_{TRIG} is opened, S_{SELF} sustains the current flow until the current lowers below I_q . Thanks to switch S_1 (whose threshold voltage is $-V_B$), the “Photon” signal effectively drives S_{TRIG} only if the reverse voltage is above breakdown, otherwise it is shunted in order to avoid unwanted ignitions out of the Geiger-mode regime.

The dependence of SPAD series resistance on the excess bias not only determines the current peak intensity, but also the quenching speed. Stray and junction capacitances affect quenching and recovery transitions and signal read-out.

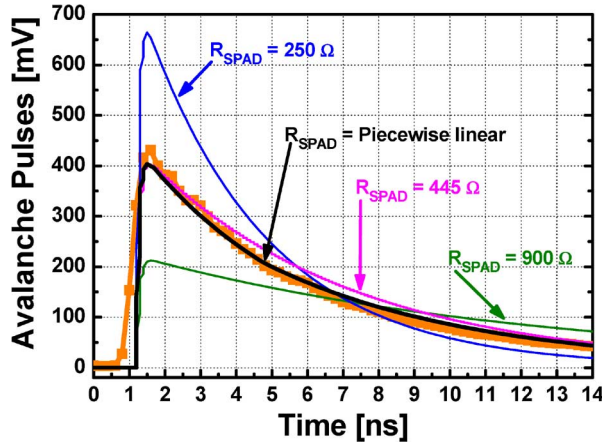


Fig. 5. Measurement (squares) versus simulations (lines) of the avalanche voltage pulse across R_S , at various R_{SPAD} series resistances. The SPAD [4] was operated with the PQC of Fig. 1.

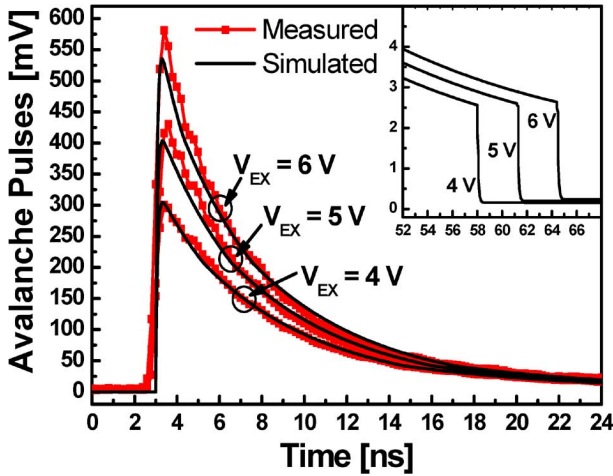


Fig. 6. Measurements (dots) versus simulations (lines) of the avalanche voltage pulse across R_S of Fig. 1, at different excess biases V_{EX} . The inset shows the current self-quenching when I_D gets lower than $100 \mu A$.

IV. EXPERIMENTAL VALIDATION

We crosschecked simulations and measurements with various SPADs. In the following, we report a $50\text{-}\mu\text{m}$ diameter silicon SPAD [4], with $V_{D1} = V_B = 21.54 \text{ V}$, $V_{D2} = 21.9 \text{ V}$, $I_{D2} = 250 \mu A$, $V_{D3} = 22.4 \text{ V}$, $I_{D3} = 1 \text{ mA}$, $V_{D4} = 24.2 \text{ V}$, $I_{D4} = 5 \text{ mA}$, $V_{D5} = 26.0 \text{ V}$, $I_{D5} = 10 \text{ mA}$, $V_{D6} = 27.0 \text{ V}$, and $I_{D6} = 14 \text{ mA}$. Such breakdown voltage and (V_{Di}, I_{Di}) points were extracted from the experimental I - V characteristics of the SPAD, shown in Fig. 3, as measured by a curve tracer. Stray and junction capacitances were measured with a standard C-meter. We used the simple PQC circuit of Fig. 1, with $R_S = 50 \Omega$ and $R_L = 220 \text{ k}\Omega$, with the addition of a $C_L = 10 \text{ pF}$ from cathode to ground, for improving the current read-out through R_S [1].

The importance of accurately modeling the SPAD series resistance R_{SPAD} is proved by Fig. 5, which shows the avalanche pulse across R_S . Too low a resistance gives a high current peak

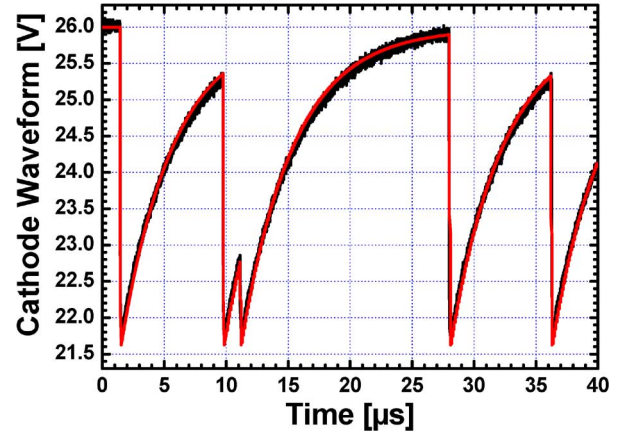


Fig. 7. Measurement (dots) versus simulation (line) of the cathode waveform of the SPAD [4] operated with the PQC of Fig. 1, in case of a photon burst, with a mean rate of 125 kcounts/s .

and short quenching time constant. Even with a value chosen to fit the peak amplitude (445Ω in Fig. 5), the time constant is hardly matched. Instead, the proposed model guarantees a good matching of the whole waveform (Fig. 5).

Fig. 6 reports three comparisons between simulations and measurements of the avalanche pulse, at different excess biases. Thanks to the nonlinear resistor implemented into the model, a very good matching is obtained, both in terms of avalanche peak amplitude and pulse waveform. Note also the current self-quenching, at a voltage $I_q \cdot R_S$ (see inset of Fig. 6).

Fig. 7 shows the simulated and measured cathode voltage, at an average photon flux of about 125 kcounts/s . Note the correctly simulated recovery transitions after each retriggerings due to the photon burst.

V. CONCLUSION

We presented an enhanced PSpice circuit model for SPADs operated in Geiger-mode, able to accurately predict the avalanche current static and dynamic behaviors, during ignition, self-sustaining, self-quenching, and recovery transitions, with both single-photon and burst triggerings. Detailed comparisons between simulations and measurements proved the model consistency and accuracy.

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